

myCompiler

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FILES +

main.m

Blocks.ai

Reach your agent anywhere. No inbound ports, no DNS, no static IP, any device, behind NAT. **Blocks.ai** →

blocks.ai

Ads by EthicalAds

```
1 clear;
2 clc;
3 close all;
4
5 %% 5G NR Parameters
6 BW = 100e6;           % Bandwidth (100 MHz)
7 SCS = 30e3;           % Subcarrier spacing (30 kHz)
8 Nsub = floor(BW/SCS); % Number of subcarriers
9
10 codeRate = 0.85;      % Coding rate
11 MIMO = 4;             % 4x4 MIMO
12 OFDMSymbols = 14000; % Approximate OFDM symbols/sec (30 kHz SCS)
13
14 %% Modulation Schemes
15 mod_scheme = {'QPSK','16QAM','64QAM','256QAM'};
16 bits = [2 4 6 8];
17
18 throughput = zeros(1,length(bits));
19
20 %% Throughput Calculation
21 for i = 1:length(bits)
22
23     throughput(i) = bits(i) * Nsub * OFDMSymbols * ...
24                     codeRate * MIMO / 1e6; % Mbps
25
26 end
27
28 %% Display Results
29 disp('5G NR Throughput')
30
31 for i = 1:length(mod_scheme)
32     fprintf('%s = %.2f Mbps\n',mod_scheme{i},throughput(i));
33 end
34
35 %% Plot
36 figure;
```

Run Save Output Input

5G NR Throughput
QPSK = 317.30 Mbps
16QAM = 634.60 Mbps
64QAM = 951.90 Mbps
256QAM = 1269.21 Mbps

[Execution complete with exit code 0]

5G NR Throughput

Modulation Scheme	Throughput (Mbps)
QPSK	317.30
16QAM	634.60
64QAM	951.90
256QAM	1269.21